Fitting Tree Cown Width Models

A Test Using a Large Crown Width Data Set (Working Draft)

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1.0 Introduction

The Forest Vegetation Simulator (FVS, Wykoff and others 1982) currently has variants that cover the continental United States (CONUS). While the variants are implemented under the FVS umbrella that standardizes the inputs and outputs, the equations and architecture of the individual variant models can be different. The work reported here is part of a larger project that is attempting to build a continental United States (CONUS) tree-list growth model that has consistent species level models.

Competition is an important input for predicting tree growth. One measure of competition that has been widely used in crown competition factor (CCF, Krajicek and others 1961) which is a relative measure that is scaled so that a value of 100 indicates (in general) that tree crowns are just touching. A tree's CCF is predicted using open grown (or maximum crown) width (MCW) equations that are functions of DBH. Historically these have been simple linear or in some cases quadratic functions as follows:

$$MCW = A + B * DBH$$
$$MCW = A + B * DBH + C * DBH^2$$

where DBH is the diameter at breast height (inches) and MCW is the maximum crown width (feet) for open grown trees. The CCF for a tree is the maximum crown area

$$CCF_{tree} = MCA_{tree} = pi * (MCW^2)/(4 * (43560/100) = 0.00180303 * MCW^2$$

The per acre CCF is computed by summing the CCFtree values multiplied by their per acre expansion factors.

A possible advantage of CCF over other competition indices is that it is based on individual species MCW values.

A challenge of modeling MCW is finding open-grown trees, especially those with large diameters and some minor species. Typically MCW models have been linear or quadratic. Applying linear type models is a problem since they keep increasing with DBH while quadratic model forms typically peak and then decrease. Because it is difficult to find very large open-grown trees that means to apply these models extrapolations are typically needed and constraints are need so predictions are reasonable. There have been some MCW models fit with model forms that curve (Russell and Weiskittel 2010) and therefore reduce the problems associated with extrapolation, however, these are a minority of the avaible equations.

Another crown width equation that is use in tree-list models is forest-growth or largest crown width (LCW) (e.g. Hann 1997 and Bechtold 2003 and 2004) . These are used to estimates crown closure or as a base for models that estimate crown shape (Hann 1999).

Using equations for MCW and LCW that are derived using different equations and different data set has the possibility of causing inconsistencies between the two estimates.

This report investigates using a power function to estimates MCW and LCW) using the Nation Forest Health Monitoring data set as used by Russell and Weiskitttel (2010) in Maine. If this is possible it would make it possible to use more well behaved functions and model MCW for many species.

2.0 Methods

The National Forest Health Monitoring (FHM) data will be used (Mangold 1998). Russell and Weiskitttel (2010) utilized this data from Maine with two other data sets to fit MCW and LCW models. This data set uses Forest Inventory and Analysis (FIA) procedures with a focus on crown characteristics, including crown width. In addition to diameter at breast height (DBH, inches) and crown ratio (CR), the width of the widest part of the tree crown (CROWN_DIAMETER_WIDE, CWwide, feet) and the width at 90-degrees (CROWN_DIAMETER_90, CW90, feet) will be used here. The crown width is computed as $ CW = sqrt{CWwide*CW90} $.

For this exploration the data for live Douglas-fir will be used. This is a large data set with 7126 trees that is expected to have dominant crown class trees within the upper canopy of plots and we felt has the best chance of succeeding. The trees for the full Douglas-fir data set is summarized below.

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   5.000   7.000   9.843  11.772  14.330  68.661
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.00   45.00   65.00   63.86   85.00   99.00
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.00   11.96   15.00   16.24   19.35   65.18
```

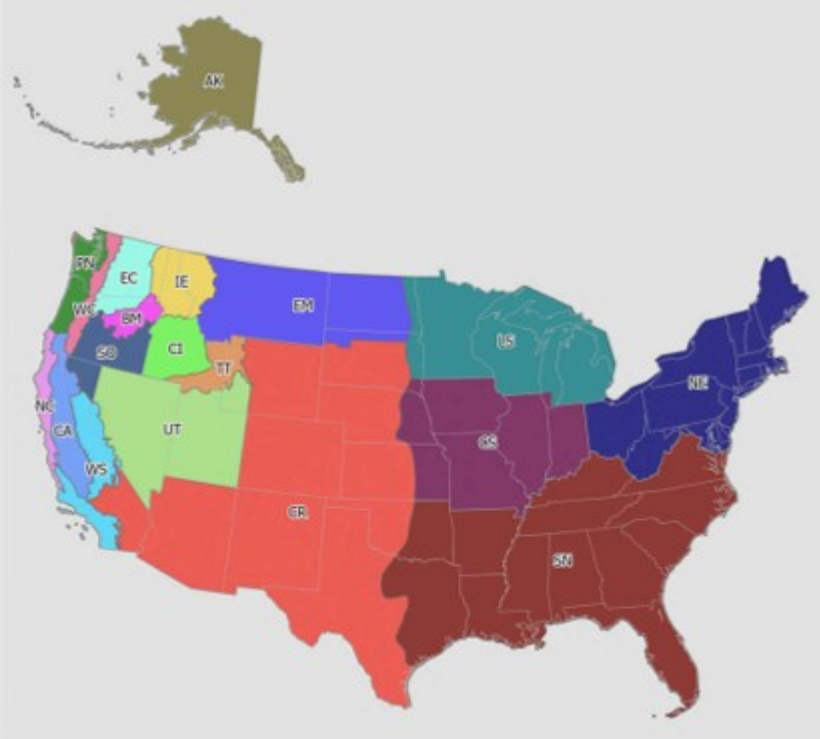
The MCW will be fit using the R packages "quantreg" using the upper 95th percentile of the crown width data (95th percentile was arbitrarily used for this test). The LCW equations were fit with "nls".

Fits were done with R Core Team (2025) version 4.5.1.

3.0 Results

3.1 FVS MCW Models

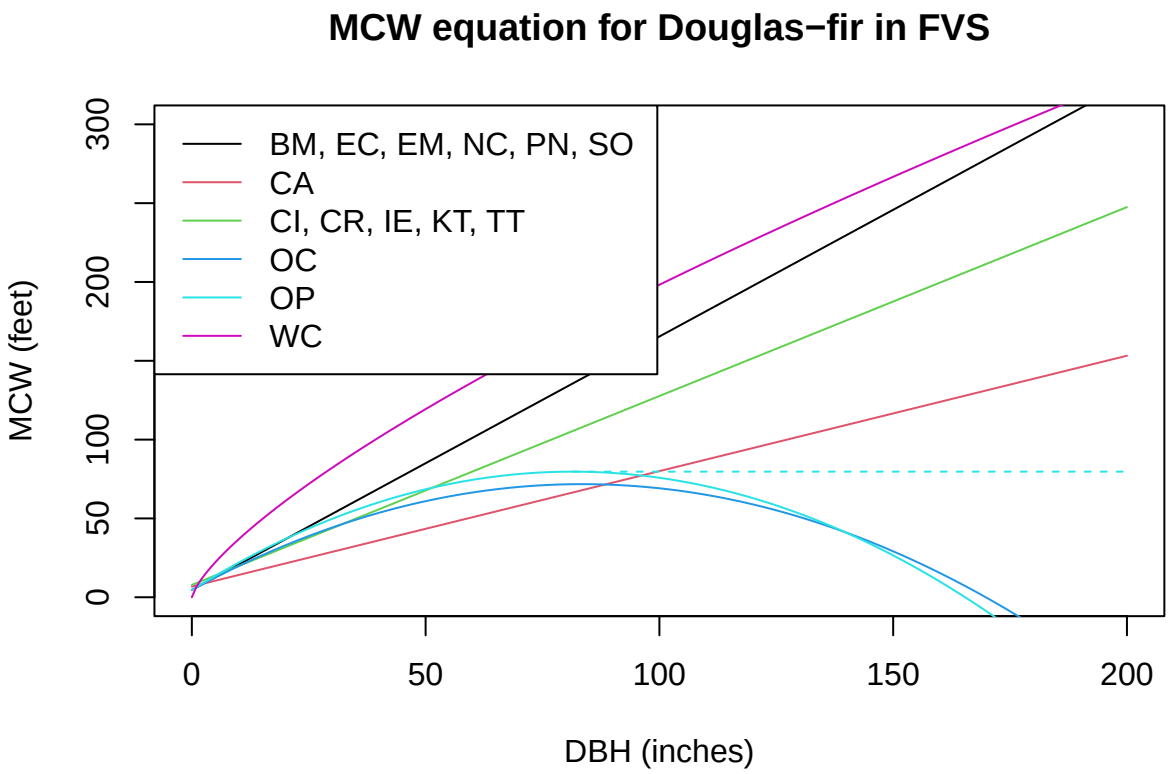
The variant map below does not include three variants that overlap other variants: KT (is the northwestern corner of Montana and is included in the IE variant), OP (is ORGANON SMC/NWO variant and is similar to the PN variant), and OS (is the SW Oregon version of ORGANON and is in SW Oregon and would overlap the northern portions of the NC and CA variants and the southern portion of the WC variant).



The graph below shows the six MCW equations that represent Douglas-fir and in FVS variants. All of the FVS Douglas-fir MCW equations are linear or quadratic except the equation "WC" (west cascades) which is power function.

Table 1: FVS variants MCW equations for Douglas-fir.

EqnNo	Variants	A1	A2	A3	maxDBH
1	BM, EC, EM, NC, PN, SO	4.639	1.608	0	NA
2	CA	6.81	0.732	0	NA
3	CI, CR, IE, KT, TT	7.811	1.199	0	NA
4	OC	4.637	1.608	-0.00963	NA
5	OP	4.62	1.843	-0.01131	81.45
6	WC	6.81	0.732	0	NA



The graph is extended to a 200-inch DBH which is the largest Douglas-fir DBH in the FIA tree data set. This means that the FVS MCW equations would likely need to extrapolate. For example, the data of Paine and Hann (1982) Douglas-fir equation for southwest Oregon when up to about 57-inch DBH.

There are a couple interesting things to note:

1. The PN (linear MCW model) and OP (quadratic MCW model) variants both cover coastal Oregon and Washington and are similar until the model types diverge.
2. The CA (north central) and NC (north coast) variants are very different curves.
3. EM (eastern MT) and CR (central Rockies) are more similar but very different from CA.

This begs the question of how much this has to do with data and how much is due to growing conditions. Paine and Hann (1982) suggest that there are difference in MCW with latitude across southwest OR.

3.2 Fitting MCW Models

The first model to fit is a typical MCW quadratic equation.

- Model 1: MCW = B1 + B2*DBH
- Model 2: MCW = B1 + B2DBH + $B3$DBHˆ2
- Model 3: MCW = B1*DBHˆB2

Models 1 and 2 are typical MCW equations that has been used for many species (e.g., Paine and Hann 1982). A potential issue with these models is how they extrapolate. The linear model (1) continues to increase and

can predict unreasonbly large CW values as DBH increases. The quadratic model (2) can peak and decrease if B2 < 0. The third model is a power function (Russell and Weiskitttel 2010) that does not peak and does not continue to increase as a linear function would.

```
##
## Call: rq(formula = CW ~ DBH, tau = 0.95, data = temp)
##
## tau: [1] 0.95
##
## Coefficients:
##             Value    Std. Error t value  Pr(>|t|)
## (Intercept) 10.69994  0.26370   40.57675  0.00000
## DBH          1.04177  0.02640   39.45498  0.00000

## [1] 46192.96
## attr(,"edf")
## [1] 2

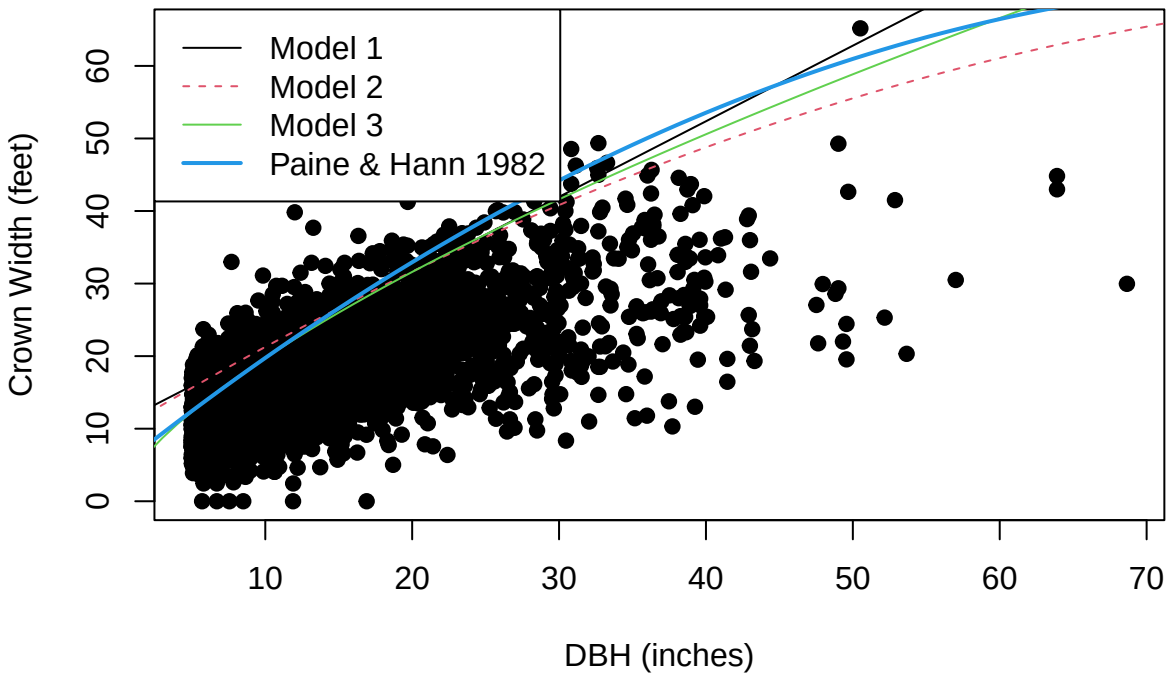
##
## Call: rq(formula = CW ~ DBH + I(DBH^2), tau = 0.95, data = temp)
##
## tau: [1] 0.95
##
## Coefficients:
##             Value    Std. Error t value  Pr(>|t|)
## (Intercept)  9.68078  0.82959   11.66941  0.00000
## DBH          1.21928  0.14545    8.38280  0.00000
## I(DBH^2)    -0.00605  0.00547   -1.10530  0.26907

## [1] 46173.25
## attr(,"edf")
## [1] 3

##
## Call: nlrq(formula = CW ~ A1 * DBH^A2, data = temp, start = c(A1 = 1.37,
##     A2 = 0.572), tau = 0.95, control = list(maxiter = 100, k = 2,
##     InitialStepSize = 1, big = 100000000000000000000, eps = 0.0000001,
##     beta = 0.97), trace = FALSE)
##
## tau: [1] 0.95
##
## Coefficients:
##    Value    Std. Error t value  Pr(>|t|)
## A1  4.14090  0.13203   31.36309  0.00000
## A2  0.67842  0.01297   52.28841  0.00000

## [1] 48181.44
## attr(,"edf")
## [1] 2
```

The squared term on model (2) is not significant, which is not uncommon. The power function of model 3 has the lowest AIC (whatever that means for this quantile regression). Below the three models are compared to the Douglas-fir MCW equation for southwestern Oregon from Paine and Hann (1982) as a reference.

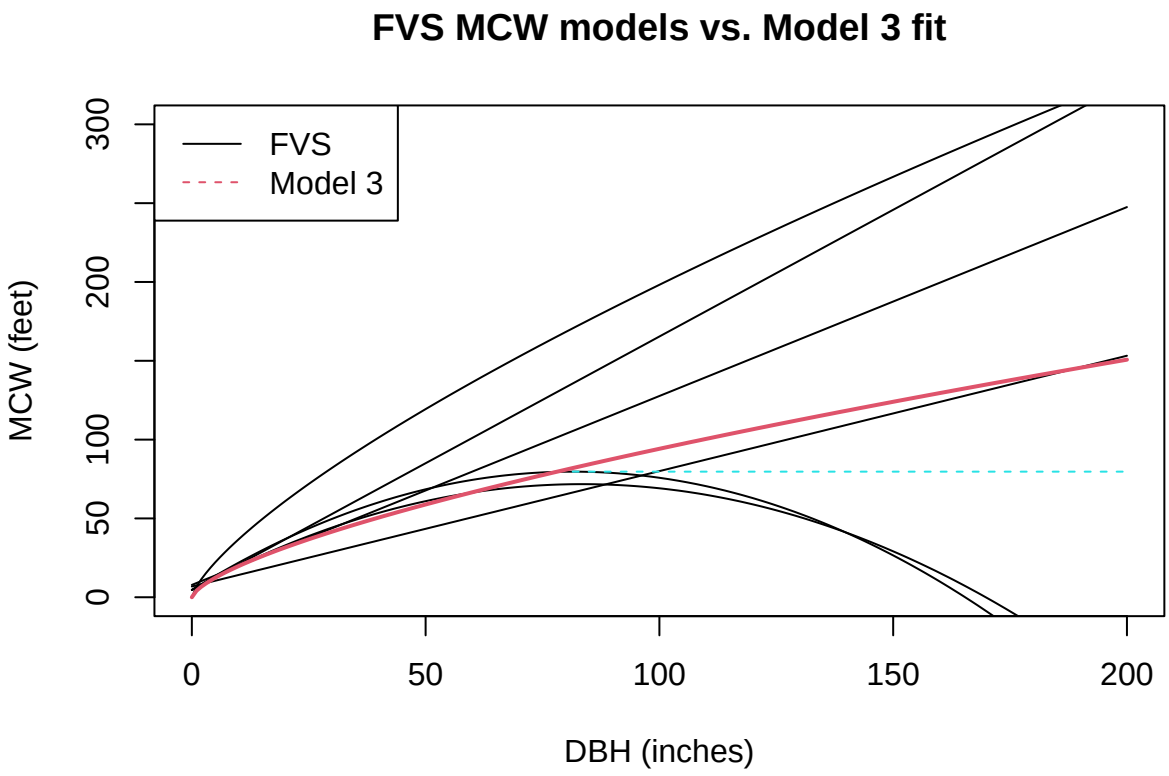


This exploratory work seems to confirm Russell and Weiskitttel (2010) results for 15 species in Maine.

The graph below plots the model 3 fit against the Douglas-fir MCW equations used for Douglas-fir in FVS.-

Table 2: FVS variants MCW equations for Douglas-fir.

EqnNo	Variants	A1	A2	A3	maxDBH
1	BM, EC, EM, NC, PN, SO	4.639	1.608	0	NA
2	CA	6.81	0.732	0	NA
3	CI, CR, IE, KT, TT	7.811	1.199	0	NA
4	OC	4.637	1.608	-0.00963	NA
5	OP	4.62	1.843	-0.01131	81.45
6	WC	6.81	0.732	0	NA



This graph is a bit discouraging given the size of some of the (extrapolated) crown widths.

3.3 Fit LCW Models

For comparison we will also fit the following LCW equations.

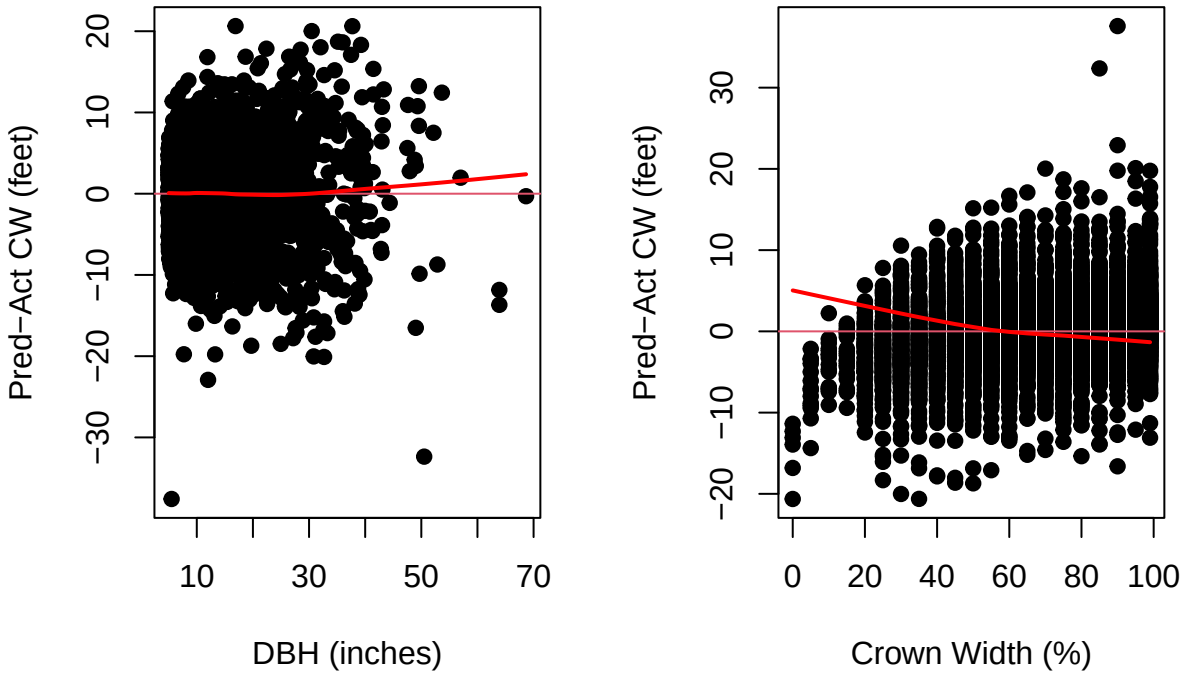
- Model 4: MCW = B1 + B2DBH + $B3$DBH^2
- Model 5: MCW = B1 + B2DBH + $B3$DBH^2 + B4*CR
- Model 6: MCW = B1*DBH^B2

Models 4 and 5 are given by Bechtold (2003, 2004). Model 6 is used by Russell and Weiskitttel 2010).

```
##
## Call:
## lm(formula = CW ~ DBH + I(DBH^2), data = temp)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -20.631  -2.510  -0.012   2.445  37.578
##
## Coefficients:
##               Estimate Std. Error t value            Pr(>|t|)
## (Intercept)  5.6655401  0.1749349   32.39 <0.0000000000000002 ***
## DBH          1.0606019  0.0220522   48.09 <0.0000000000000002 ***
## I(DBH^2)    -0.0103581  0.0005541  -18.69 <0.0000000000000002 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.284 on 7123 degrees of freedom
## Multiple R-squared:  0.5395, Adjusted R-squared:  0.5394
## F-statistic:  4173 on 2 and 7123 DF,  p-value: < 0.00000000000000022

## [1] 40963.72
```



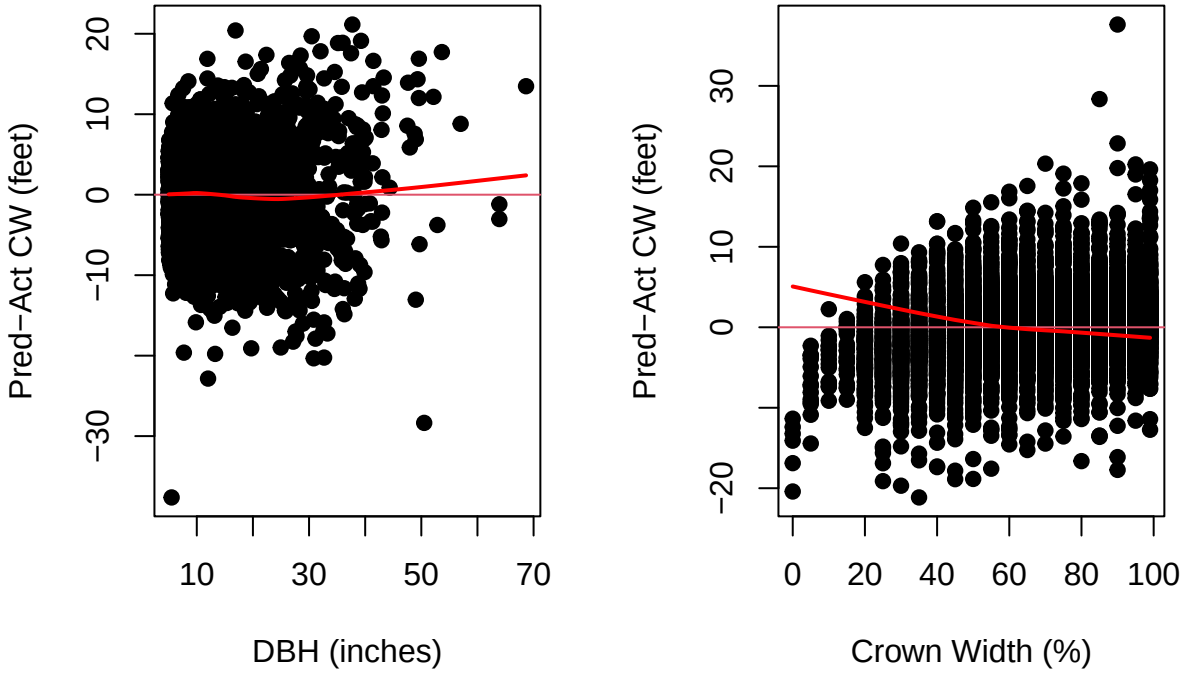

```
##
## Call:
## lm(formula = CW ~ DBH + I(DBH^2) + CROWN_RATIO, data = temp)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -19.038  -2.342  -0.038   2.259  36.212
##
## Coefficients:
##              Estimate Std. Error t value            Pr(>|t|)
## (Intercept)  2.132563   0.218314   9.768 <0.0000000000000002 ***
## DBH          1.074260   0.021135  50.829 <0.0000000000000002 ***
## I(DBH^2)    -0.010670   0.000531 -20.095 <0.0000000000000002 ***
## CROWN_RATIO  0.053709   0.002127  25.255 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 4.105 on 7122 degrees of freedom
## Multiple R-squared:  0.5774, Adjusted R-squared:  0.5772
## F-statistic:  3243 on 3 and 7122 DF,  p-value: < 0.00000000000000022

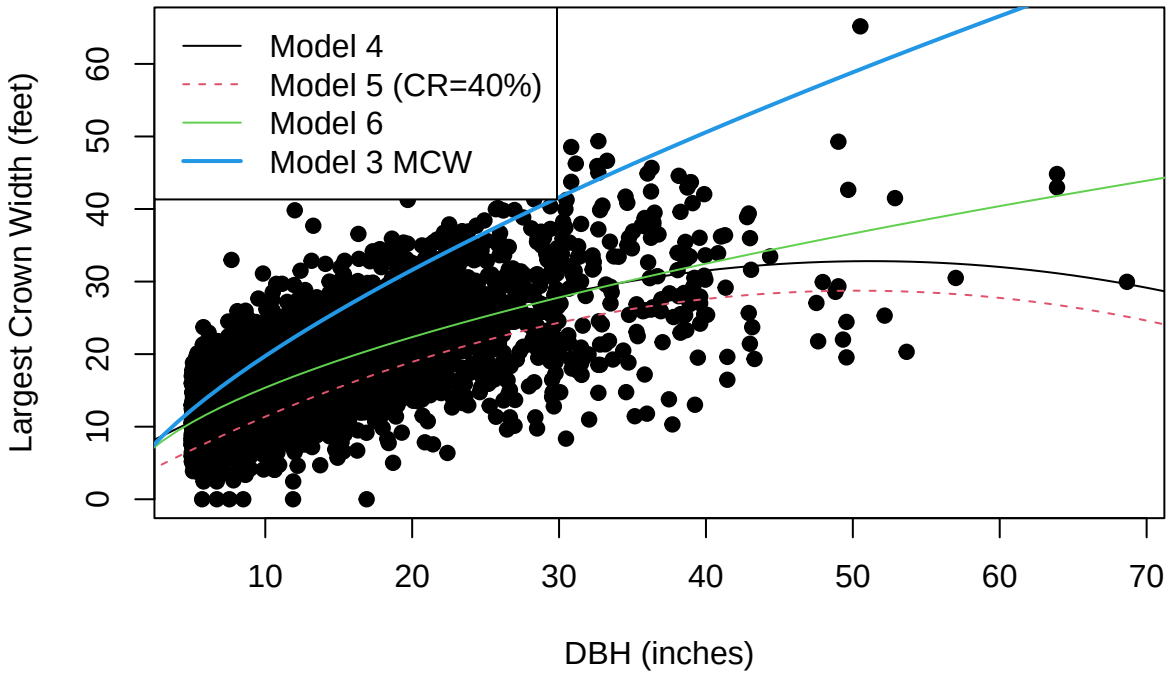
## [1] 40354.53

##
## Formula: CW ~ A1 * DBH^A2
##
## Parameters:
##    Estimate Std. Error t value            Pr(>|t|)
## A1 4.445455   0.066629   66.72 <0.0000000000000002 ***
## A2 0.539028   0.005595   96.34 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.298 on 7124 degrees of freedom
##
## Number of iterations to convergence: 4
## Achieved convergence tolerance: 0.000001578

## [1] 41007.5
```



These are plotted below with the MCW model (3).



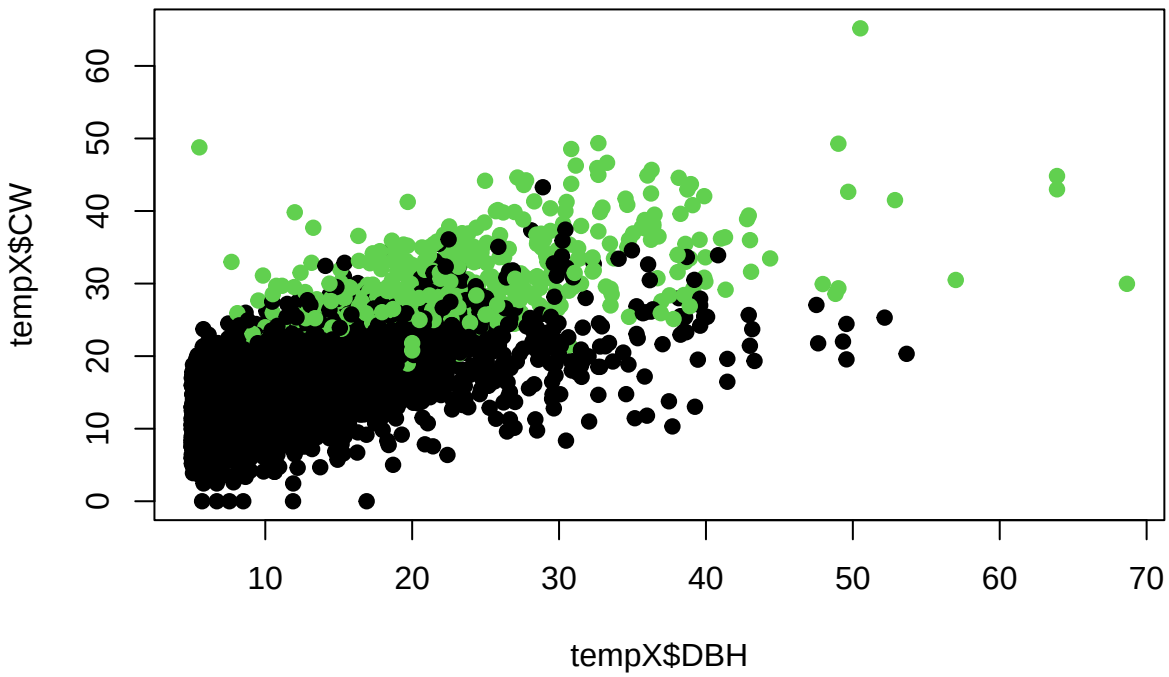
The model fits are similar in fits and residuals through the range of the data. The AIC was about 1.5% less with the addition of CR to model 2. The AIC for Model 6 was about 1.1% larger than Model 4. Model 6 does not peak and a decreasing proportion to the MCW fit (Model 3) which is a more appealing behavior than the quadratic model form.

4.0 Discussion

This test confirms the work of Russell and Weiskitttel (2010) that the NHM data is a potential source of data for fitting MCW equations, in addition to LCW models (also Bechtold 2003, 2004) using a major west coast species. This also confirms an advantage of using a power function model forms as in Russell and Weiskitttel (2010).

Douglas-fir was a large data set and with large trees expected in the upper crown classes (dominants). A question remains on how this approach would work with smaller data sets and species that are found in mixed species stands and are not abundant in the dominant crown class (less tolerant species)? This may not be a big issues, but how could these be identified? Russell Weiskittel (2010) included fits for 7 conifer and 8 hardwoods species in Maine which suggests this approach could work for a wide range of species.

One idea to decide if a species might be a candidate for this analysis (i.e. have trees that likely characterize MCW) would be to assign the CW percentile of each trees (by stand/plot) and then plot just the species of interest. For example the plot below shows the percentile of each tree's CW within each stand with all trees. The plot is just Douglas-fir and the green indicates those Douglas-fir trees in the 95th percentile of all tree crown widths. If there was no or little green, then the species may not have large crowns and indicate mid- or under-story type trees.



The test done here used quantile regression with the 95th percentile. The 95th percentile was an arbitrary choose and it is possible that something else might be better (it does get harder to fit as the percentile gets closer to 100). In some cases it might be possible that some things like Stochastic Frontier Analysis might also work. Other than fit statistics it is not clear what the best way to see if the estimates of MCW are reasonable since we are using an upper part of the data distribution. For some species there are published MCW curves to compare with (as done here), but for other species there is not (especially minor species). The champion tree data base does have crown width, but these may be large old trees and not necessarily open grown (they are champions by and combined index of circumference, height and crown width).

For MCW, one wonders how much of the differences we see in the variants of FVS are due to within data set difference (data distributions, collection definitions and leverage points) and maybe model form rather than location? Open-grown is a specific definition and getting "good" trees requires effort and miles of searching. It seems reasonable that location factors that impact productivity and tree growth might play a part (Paine and Hann 1982 miles south (LAT), Bechtold 2003, 2004 (LAT/LONG/ELEV)) but relatively how much?

These results (and the results of Russell Weiskittel 2010) also makes one wonder how much difference there is in CW for large dominant trees and open-grown trees? These results suggest maybe not much (at least for how open-grown might be defined).

This work did not consider if CR variables contributed to the predictions of LCW using the power function. Both Hann (1997) and Bechtold (2003 ad 2004) scaled LCW using CR so it might be worth considering. Unfortunately the FHM data set does not include total height (no crown lengths or HT:DBH ratios).

One possible concern with the power function for MCW and LCW models is that it predicts a CW of 0 for a DBH of 0 which might give too small of CW predictions for small trees. Again, this may not matter, but should be considered. A possible alternative model form would be:

- CW = B0 + B1*DBHˆB2

The challenge would be to get reasonable estimate of the B0 parameter (the crown with of small trees) since the FHM data set just measures trees to 1-inch and may be heavily influenced by larger trees. As a test this model was fit for MCW and LCW. The B0 estimates were significant and 5.6 feet for MCW and -4.4 feet for LCW.

5.0 References

Bechtold, W.A. 2003. Crown-diameter prediction models for 87 species of stand-grown trees in the eastern United States. Southern Journal of Applied Forestry. 27(4): 269-278.

Bechtold, W.A. 2004. Largest-crown-width prediction models for 53 species in the western United States. Western Journal of Applied Forestry 19(4):245-251.

Hann, D.W. 1997. Equations for predicting the largest crown width of stand-grown trees in western Oregon. Research Contribution 17. Forestry Research Laboratory, Oregon State University, Covallis. 14 p.

Hann, D.W. 1999. An Adjustable Predictor of Crown Profile for Stand-Grown Douglas-Fir Trees. Forest Science . 45(2):217-225.

Mangold, R.D. 1998. Overview of the Forest Health Monitoring Program. In: Integrated Tools For Natural Resource Inventories in the 21st Century, M. Hansen and T. Burk, eds., August 16-20, 1998. Boise, ID. Pages 129-140.

Paine, D.P.; Hann, D.W. 1982. Maximum crown-width equations for southwestern Oregon tree species. Research Paper 46. Forestry Research Laboratory, Oregon State University, Covallis. 20 p.

R Core Team (2025). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria. https://www.R-project.org/.

Russell, M.B.; Weiskittel, A.R. 2010. Estimating tree crown widths for the primary Acadian species in Maine. Joint meeting of the Forest Inventory and Analysis (FIA) symposium and the Southern Mensurationists. Pages 45-52.

Wykoff, William R.; Crookston, Nicholas L.; Stage, Albert R. 1982. User's Guide to the stand prognosis model. Gen. Tech. Report INT-133. Ogden, UT: U.S. Department of Agriculture, Intermountain Forest and Range Experiment Station. 112 p.

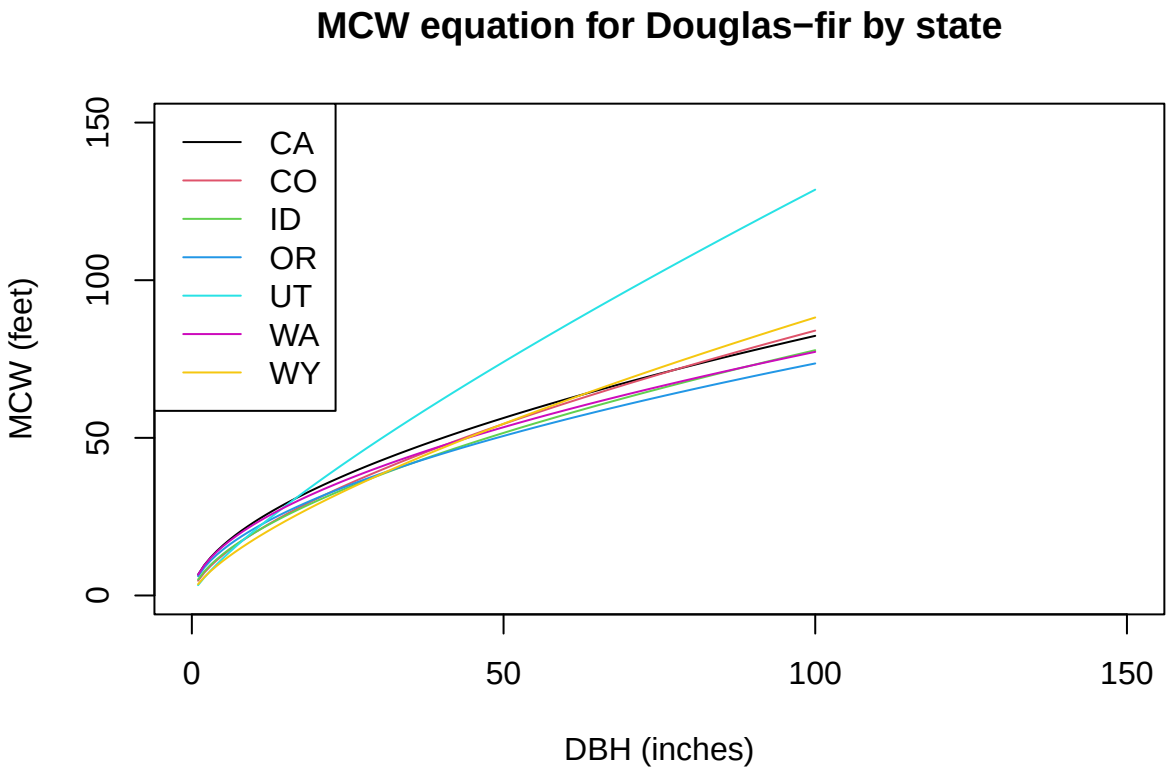
Appendix

Just for fun, lets fit the MCW for seven states with Douglas-fir using the same procedures as above in section 3.2.

Table 3: Summary for FHM Douglas-fir DBH measurements by state.

STATE	N0bs	Mean	Min	Max	STATENM
6	880	11.44	5	52.87	CA
8	510	9.98	5	28.7	CO
16	1566	11.71	5	63.9	ID
41	1825	13.41	5	68.66	OR
49	194	10.51	5	33.4	UT
53	1992	11.29	5	53.66	WA
56	159	8.736	5	20.5	WY

```
## [1] 1
## [1] 2
## [1] 3
## [1] 4
## [1] 5
## [1] 6
## [1] 7
```



This mildly interesting given that all the states except 49 (UT) cluster together which has the 4th almost trees with 900 (it is interesting to not that if the regression used tau = 0.90 WY would jump up out of the cluster and be more like UT which might suggest a sensitivity of tau). This may also be a data distribution issue with large DBH and CW trees. This also does not give any indication that we see any difference between CA and the more norther OR and WA MCW estimates. While the equation type matters (linear is especially problematic), the question remains open, data or environment (or both)?